Calculus I

Computation problems for training #1

Topics:

- Functions and their graph
- Combining functions
- Trigonometrical functions

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Problem 1. An airplane takes off from an airport and lands an hour later at another airport, 400 miles away. If t represents the time in minutes since the plane has left the terminal building. Let $x(t)$ be the horizontal distance traveled and $y(t)$ be the altitude of the plane.

a) Sketch a possible graph of $x(t)$.

b) Sketch a possible graph of $y(t)$.

c) Sketch a possible graph of the ground speed.

d) Sketch a possible graph of the vertical velocity.

Problem 2. The number N (in millions) of US cellular phone subscribers is shown in the table. (Midyear estimates are given.)

t	1996	1998	2000	2002	2004	2006
N	44	69	109	141	182	233

a) Use the data to sketch a rough graph of N as a function of t.

b) Use your graph to estimate the number of cell-phone subscribers at midyear in 2001 and 2005.

Problem 3. A stone is dropped into a lake, creating a circular ripple that travels outward at a speed of $60\,cm/s$.

a. Express the radius r of this circle as a function of the time t (in seconds).

b. If A is the area of this circle as a function of the radius, find $A \circ r$ and interpret it.

Problem 4. If $u(x) = 4x - 5$, $v(x) = x^2$, and $f(x) = 1/x$, find formulas for the following.

a. $u(v(f(x)))$ **b.** $u(f(v(x)))$
c. $v(u(f(x)))$ **d.** $v(f(u(x)))$
e. $f(u(v(x)))$ **f.** $f(v(u(x)))$

Problem 5. Let $f(x) = x - 3$, $g(x) = \sqrt{x}$, $h(x) = x^3$ and $j(x) = 2x$. Express each of the following functions as a composite involving one or more of f, g, h, and j.

a. $y = \sqrt{x} - 3$ **b.** $y = 2\sqrt{x}$
c. $y = x^{1/4}$ **d.** $y = 4x$
e. $y = \sqrt{(x-3)^3}$ **f.** $y = (2x - 6)^3$.

Problem 6. Assume that f is an even function, g is an odd function, and both f and g are defined on the entire real line $\mathbb{R}$. Which of the following (where defined) are even? odd?

a. fg **b.** f/g **c.** g/f
d. $f^2 = ff$ **e.** $g^2 = gg$ **f.** $f \circ g$
g. $g \circ f$ **h.** $f \circ f$ **i.** $g \circ g$

In the following exercise, one of $\sin(x)$, $\cos(x)$ and $\tan(x)$ is given. Find the other two if x lies in the specified interval.

Problem 7. $\sin(x) = \frac{3}{5}, \quad x \in \left[\frac{\pi}{2}, \pi\right]$.

Problem 8. $\cos(x) = \frac{1}{3}, \quad x \in \left[-\frac{\pi}{2}, 0\right]$.

Problem 9. $\tan(x) = \frac{1}{2}, \quad x \in \left[\pi, \frac{3\pi}{2}\right]$.

Problem 10. Are there two functions f and g such that $f \circ g = g \circ f$? Justify your answer.

Problem 11. Suppose $f(x)$ is odd. Is $g(x) = f(x) - 2$ also odd? What if f is even instead? Justify your answer.

Problem 12. Show that if f is both even and odd, then $f(x) = 0$ for every x in the domain of f.